

Chapter 1 ZFC Axioms

Chapter 2 Construction of Real Numbers

2.1 Natural Numbers

2.2 Integer Numbers

2.3 Rational Numbers

2.4 Real Numbers

† **Axiom** We could construct real number $(\mathbb{R}, +, \times, <)$ as following:

- Field Axioms:
- Order Field Axioms:
- Archimede Field:

2.4.1 Completeness Condition

♠ **Proposition 2.4.1.1** The followings are equivalent:

- Metric space $(\mathbb{R}, | - |)$ is complete;
- For any upper subset $M \subseteq \mathbb{R}$, $\sup(M)$ exists;

Cauchy Sequence

◇ **Lemma 2.4.1.2** Suppose $M \subseteq \mathbb{R}$ be upper bounded, then $\sup(M)$ exists.

◦ **Proof** Select upper bound N and $x \in M$,

$$a_0 := x, b_0 = N, c_n := \frac{a_n + b_n}{2} \quad a_{n+1} := \begin{cases} a_n & \forall x \in M c_n \geq x \\ c_n & \text{Elsewise} \end{cases} \quad b_{n+1} = \begin{cases} b_n & \exists x \in M c_n < x \\ c_n & \text{Elsewise} \end{cases}$$

$(a_n), (b_n)$ are monotone.

Then, $|a_n - b_n| \leq 2^{-(n-1)}|a_1 - b_1| \Rightarrow |b_{n+m} - b_n| \leq |b_n - a_n| \leq 2^{-(n-1)}|b_1 - a_1|$. It is Cauchy sequence.
Thus $b_n \rightarrow z \in \mathbb{R}$. And $a_n \rightarrow z$:
 z is $\sup(M)$. Suppose $y \in M$, we have

$$y > z \Leftrightarrow \exists N_0 (n > N_0 \rightarrow y > a_n)$$

Contradict to the definition. □

Chapter 3 title